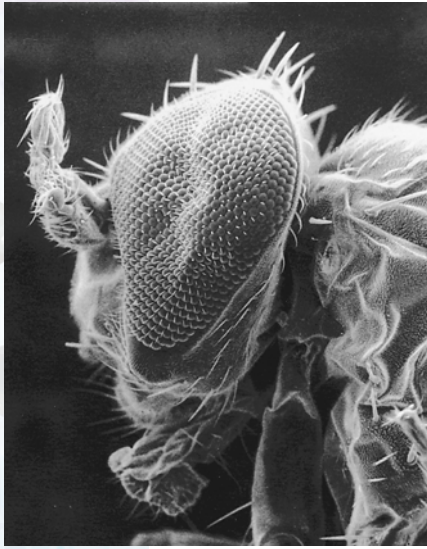




In place of its antennae this fly grew legs. Normally expressed in the thoracic region of a fly, the homeotic selector gene *antennapedia* “kick starts” a series of events leading to leg growth. A mutated form of this gene allowed for its expression in the head region of the fly.

genes, it sets off a cascade of events that results in the creation of legs. Remarkably, however, this single gene is sufficient by itself to turn on the leg-producing program, and its absence keeps the program silent. It can even turn it on in other segments. For instance, when *antennapedia* is mutated to allow it to be expressed in head segments, a pair of legs develops in place of the normal appendages, antennae (hence the gene name, which means “antenna foot”).

Intriguingly, the sequence of homeotic selector genes along the fly chromosome matches the order of segments in which each is expressed. That is, the genes expressed in head segments come first, followed by those expressed in thoracic segments, then the abdomen, then the tail. The way in which this correspondence is exploited during development is still unknown, but the arrangement is clearly not accidental. Related genes have been found in vertebrates, including humans, and the same pattern holds: Genes expressed more anteriorly precede those expressed toward the posterior.

Homeotic Genes in Other Species

Homeotic genes also control development in other species, from yeast to humans, although the details are not as clear as they are for the fruit fly. Homeotic genes, called Hox genes, control the development of segments in the mammalian hindbrain, for example, and help establish the anterior-posterior and dorsal-ventral axes in the limbs. Vertebrates have duplicate copies of the Hox genes on several chromosomes, all of which function together to specify, for example, limb development. The multiple copies provide a redundancy not found in the fruit fly, thus making the effect of individual genes harder to detect. Nonetheless, by “knocking out” multiple versions of a particular Hox gene, researchers have shown their dramatic effects. For example, mice that are missing two copies of Hox 11 have no forelimbs, and the wrist is fused directly to the elbow.

The Homeobox

As transcription factors, the homeotic gene products must bind to DNA. Sequence analysis of both gene and protein has revealed that all share a 180-nucleotide DNA stretch, termed the homeobox, the sequence of which has remained almost unchanged over many millions of years of evolution. It codes for a 60-amino acid long DNA binding region, called the homeodomain. The homeodomain sequence of *antennapedia* and the mouse homeotic gene Hox B-7 differ by only two amino acids, despite having diverged several hundred million years ago.

Programmed Cell Death: Apoptosis

Development of a multicellular creature requires not only cell differentiation, but in some cases, cell death. **Apoptosis** helps create the spaces between the fingers, for instance. During brain development, nerve connections are sculpted through the apoptotic death of billions of cells. In *C. elegans*, exactly 131 cells die by apoptosis.

Cells can be directed to the apoptotic pathway if they fail to receive appropriate signals from their neighbors. In this way, it is thought that cells in the wrong location—a bone cell in the gut, for instance—might

apoptosis programmed cell death

be terminated to prevent damage to the organism. The death program itself is carried out within the cell by activation of specific genes that ultimately trigger proteases, which are enzymes that break down cell contents, including the chromosomes. **SEE ALSO** APOPTOSIS; FRUIT FLY: *DROSOPHILA*; ROUNDWORM: *CAENORHABDITIS ELEGANS*; TRANSCRIPTION FACTORS; TRANSGENIC ORGANISMS; ZEBRAFISH.

Richard Robinson

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Diabetes

Diabetes mellitus is a group of diseases characterized by elevated levels of **glucose** in the blood. Diabetes is caused by problems producing or responding to the hormone insulin. Insulin is produced in the pancreas by specialized cells called beta cells, in response to the presence of glucose absorbed through the gastrointestinal tract following a meal. Insulin promotes the uptake of glucose into muscle and fat cells, and it promotes the storage of excess glucose in the liver.

Excess blood glucose over time damages organs, particularly the eyes, kidneys, nerves, heart, and blood vessels. It is the leading cause of adult blindness, end-stage kidney disease, and lower limb amputations, and it is a major risk factor for heart attacks and strokes. Diabetes is classified into four major groups: type 1 diabetes (T1DM), type 2 diabetes (T2DM), other specific types, and gestational diabetes (GDM), occurring during pregnancy. Approximately 5 percent to 8 percent of the people of the industrialized world have diabetes, mostly (approximately 90 percent) type 2, which at least 16 million Americans have.

Type 1 Diabetes

Type 1 diabetes is caused by beta cell destruction, leading to insulin deficiency. T1DM was previously called insulin-dependent diabetes mellitus (IDDM), because patients who have it require insulin for survival. It was also called juvenile-onset diabetes mellitus, because most type 1 diabetics are children or young adults. At the time of diagnosis, about 85 percent to 90 percent of people with type 1 diabetes have **antibodies** directed against components of their beta cells, indicating that the immune system is responsible for the progressive and irreversible beta cell destruction.

Current evidence indicates a genetic component to T1DM. HLA (histocompatibility leukocyte antigen) genes are a group of genes on chromosome 6 that encode proteins that are part of the immune system. Normally the immune system defends the body against disease by destroying foreign

glucose sugar

antibodies immune-system proteins that bind to foreign molecules

cells, but in the case of type 1 diabetes, the body's immune system destroys its own beta cells.

Certain types of HLA genes are strongly associated with type 1 diabetes, and other types protect against its development. However, these HLA genes are neither necessary nor sufficient to cause or protect from type 1 diabetes. T1DM is therefore a “complex” genetic disorder, in which several genes interact with the environment to result in the disease. Scientists are currently working to identify these other genes, as well as environmental factors (e.g., toxins and viruses) that provoke the development of T1DM.

Type 2 Diabetes

Type 2 diabetes is itself a group of disorders caused by some combination of insulin resistance—which occurs when cells' ability to *respond* to insulin is compromised—and insulin deficiency, which occurs when the beta cells' ability to *make* insulin is compromised. T2DM has, in the past, been called adult-onset diabetes, because most people with T2DM were adults. It was also called non-insulin-dependent diabetes mellitus (NIDDM), because people with type 2 diabetes usually do not require insulin injections. In the United States, T2DM is especially prevalent among certain ethnic minorities, including African Americans, Mexican Americans, Asian Americans, and Native Americans.

Obesity is a potent risk factor for T2DM. In the last thirty years, due to increased caloric intake and physical inactivity, both of which contribute to obesity, there was an explosion in the prevalence of T2DM, and it started occurring at younger ages—even in children. In addition to its association with an unhealthy lifestyle, T2DM is known to have a strong genetic component.

Scientists have been searching throughout the genome for T2DM-susceptibility genes. One such gene, *calpain 10 protease*, was identified on chromosome 2. A common variant of this gene may predispose certain individuals to T2DM; however, the true significance of this gene variant remains to be determined. In addition, several candidate genes have shown some evidence of being involved in T2DM. However, the effect of any single candidate gene variant on the risk of developing T2DM is modest. A candidate gene is a gene for which prior knowledge of its function leads researchers to assess whether chemical variation in it is associated with a disease.

As of 2002 there was no clinically available genetic test to predict the onset of type 2 diabetes, but it is anticipated that with a better understanding of the roles of various genes in T2DM, it will eventually be possible to use multiple genetic tests to identify individuals at risk for T2DM and to predict which treatments will be most helpful in specific patients. Although genetic susceptibility plays an important role in determining the risk of developing T2DM, studies have shown that the disease can often be prevented through diet, physical activity, and weight loss.

Other Specific Types of Diabetes

The third category of diabetes, containing other specific types, includes nongenetic forms as well as single-gene forms of diabetes. One group of single-gene diabetes disorders are genetic defects in beta cell function. The

most common of the genetic beta cell defects are the disorders known as MODY, or maturity onset diabetes of the young. MODY constitutes no more than 2 percent to 5 percent of all cases of diabetes. It often occurs in children and young adults and is characterized by decreased but not absent insulin production. It is inherited in an autosomal dominant manner, which means that an affected person has a 50 percent chance of passing on the disease-version of the gene with each pregnancy. Most, but not all, people receiving a MODY gene do develop diabetes.

There are at least six different genetic forms of MODY. MODY2 is caused by a mutation in a gene on chromosome 7 that makes a protein called glucokinase, which is an enzyme in beta cells that helps to provide a chemical signal needed for insulin release. The other MODYs involve mutations in genes that encode proteins called transcription factors, which allow beta cells to develop and function properly. These are hepatocyte nuclear factor 4-alpha (*HNF4-alpha*, causing MODY1, on chromosome 20), *HNF1-alpha* (causing MODY3, on chromosome 12), insulin promoter factor 1 (*IPF1*, causing MODY4, on chromosome 13), *HNF1-beta* (causing MODY5, on chromosome 17) and *NeuroD1/beta2* (causing MODY6, on chromosome 2).

A very rare genetic insulin secretion disorder is maternally inherited diabetes and deafness (MIDD), caused by changes in the DNA of the mitochondria. The **mitochondria** are the energy powerhouses of the cell and the only part of the cell to contain DNA other than the nucleus, where most DNA is contained. MIDD and other mitochondrial disorders are maternally inherited because the fertilized egg has only mitochondria derived from the mother. The clinical features of MIDD can be similar to type 2 diabetes, and the hearing loss can be mild or even undetectable, except by special tests.

Another group of rare genetic diabetes types is characterized by extreme insulin resistance, which is defined as occurring when the ability of the body's cells to respond to insulin is severely compromised. Disorders of extreme insulin resistance include type A syndrome, leprechaunism, and Rabson-Mendenhall syndrome, and they are caused by inherited defects in the gene on chromosome 19 that makes the insulin receptor, a protein that allows cells to respond to insulin. Without properly functioning insulin receptors, insulin cannot work effectively. In addition to diabetes, individuals with insulin receptor defects may also have dental, genital, skin, and growth abnormalities. Most insulin receptor gene defects manifest in an autosomal recessive manner. That is, two defective copies of the gene are required for disease expression, and couples in which each partner has one defective copy (and in which neither is therefore affected) have a 25 percent chance of having an affected child, with each pregnancy.

Familial partial lipodystrophic diabetes (FPLD) is a rare condition in which children develop an unusual fat distribution at puberty, with little or no fat on their arms, legs, and trunk. They also develop insulin-resistant diabetes. FPLD is an autosomal dominant condition caused by mutations in the *lamin A/C* gene on chromosome 1. Another rare form of lipodystrophic diabetes is congenital (i.e., present at birth) generalized lipodystrophic (CGL) diabetes, which is autosomal recessive, and in about half of cases is due to mutations in the gamma-3-like gene (*GNG3*; also called the seipin gene), on chromosome 11.

mitochondria energy-producing cell organelle



Wolfram syndrome is a rare autosomal recessive condition presenting in childhood that includes diabetes mellitus as well as other problems, including deafness and deficiency of antidiuretic hormone. Mutations in the wolframin gene on chromosome 4 are responsible for some cases, but other cases appear to be caused by a gene in a different area of chromosome 4.

Another rare autosomal recessive childhood condition, thiamine-responsive megaloblastic anemia syndrome (TRMA), consists of several features, including blood abnormalities, deafness, and diabetes. TRMA, which responds to treatment with thiamine (a form of vitamin B), is a disorder caused by mutations in the thiamine transporter gene *SLC19A2*, on chromosome 1.

Transient neonatal diabetes (TNDM) is a condition in which infants are born requiring injected insulin but are able to make sufficient insulin later in infancy. Later in childhood or in adulthood, they may again develop diabetes, which may or may not require insulin treatment. Most cases of transient neonatal diabetes appear to be caused by the inheritance of an extra copy of a region of chromosome 6 from the father.

Many known genetic disorders other than those mentioned previously are associated with an increased risk of diabetes. Among those most strongly associated are Friedreich's ataxia, cystic fibrosis, and hemochromatosis.

Gestational Diabetes Mellitus

Hormones associated with pregnancy may cause diabetes in susceptible individuals. Although the diabetes goes away after the pregnancy, individuals who have had GDM are at increased risk of developing T2DM. Currently very little is known about the genetic basis of GDM. It is possible that some of the same genes responsible for T2DM are also involved in GDM.

Genetic Susceptibility to Complications

As mentioned above, diabetes is associated with complications involving the eyes, kidneys, blood vessels, and heart. However, not all individuals with diabetes develop these complications. There is increasing evidence that there are genes other than those that increase susceptibility to developing the disease that may influence susceptibility to developing its complications. These genes are not yet identified, but they are likely to interact with other known risk factors for complications, including poor blood-sugar control and increased blood-pressure and blood-cholesterol levels. SEE ALSO COMPLEX TRAITS; DISEASE, GENETICS OF; GENE AND ENVIRONMENT; GENE DISCOVERY; IMMUNE SYSTEM GENETICS; MITOCHONDRIAL DISEASES.

Toni I. Pollin and Alan R. Shuldiner

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Internet Resources

American Diabetes Association. <<http://www.diabetes.org>>.

Joslin Diabetes Center. <<http://www.joslin.org>>.

Juvenile Diabetes Research Foundation International. <<http://www.jdrf.org>>.

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Online Mendelian Inheritance in Man. Johns Hopkins University, and National Center for Biotechnology Information. <<http://www.ncbi.nlm.nih.gov/Omim>>.

Disease, Genetics of

Genetics is believed to play a role in almost every human disease. Even for diseases traditionally described as environmental, such as tuberculosis and HIV, scientists are discovering that genetics is implicated either in the susceptibility to infection or in the severity of the disease. In some disorders a variation within a single gene is sufficient to cause disease, while in other disorders variations within a gene must interact with the environment and other genes to cause disease. The goal of human medical genetics is to identify all the genes that are involved in human disease and determine how the genes function to cause susceptibility to disease. This knowledge leads to the development of successful therapies that improve the quality of life of affected individuals and their families.

Mendelian and Complex Disorders

Geneticists typically classify genetic disorders into two main categories: Mendelian and complex disorders. Mendelian disorders, such as sickle-cell disease, cystic fibrosis, and Duchenne muscular dystrophy, are usually rare in the general population. These disorders have predictable, recognizable inheritance patterns (such as autosomal dominant and X-linked recessive), and variations in a single gene are sufficient to cause expression of the disorder. Furthermore, only individuals who carry a **mutation** in the causative gene are at risk for expressing the disorder.

In contrast, complex disorders, such as cardiovascular disease, diabetes, cancers, and psychiatric disorders, are common in the general population. In these disorders, genetics plays a significant role, but the biology of the disease is due to a tangled web of genetic and environmental interactions. Consequently, complex disorders generally do not display the distinct inheritance patterns seen in Mendelian disorders.

While the genetic variation at a single gene may contribute to the overall risk of developing a disease, it is not expected to be sufficient for expression of the disease. A well-known example of this is the association between Alzheimer's disease and the *APOE* gene. Individuals who carry the "4" **allele** of the *APOE* gene have a higher risk and earlier age-of-onset for Alzheimer's disease than those with other alleles. Furthermore, this association is dose-dependent. Individuals who have two copies of *APOE-4* are at greater risk for developing Alzheimer's disease than individuals who carry one copy of *APOE-4* and one copy of a different allele.

How Important Are Genes?

Prior to searching for the genes involved in a disease and determining how those genes work in the various tissues of the human body, there must be clear evidence that genetics is involved in the disease. Genetic analyses are ethically and financially challenging, as well as quite laborious. Geneticists use several methods to evaluate whether or not genetics plays an important role in the **etiology** of a disorder before they begin a search for a gene. Some of these methods include familial aggregation, recurrence risks, and twin studies.

Familial aggregation can be established by obtaining a thorough family history on study participants. Individuals are simply asked if they have any

mutation change in DNA sequence

allele a particular form of a gene

etiology causation of disease, or the study of causation



prevalence frequency of a disease or condition in a population

other family members who have the disease. If individuals with the disease have a higher frequency of affected relatives compared with individuals without the disease, there is evidence of familial aggregation. Although familial aggregation for a disorder is consistent with the involvement of genetics, it may also reflect the presence of a common environmental factor to which all family members have been exposed (such as pesticides or contaminated drinking water).

Relative risk ratios are another method commonly used to determine if there exists a genetic basis to a disease. For example, in cystic fibrosis, an autosomal recessive Mendelian disorder, the risk of disease in the siblings of an affected individual is 1 in 4. The **prevalence** of the disease is about 1 in 1,600 in the general population.

In 1990 Neil Risch demonstrated that by comparing the risk of a disease occurring in the relatives of an affected individual with the risk of the disease occurring in the general population, one could measure the significance of the genetic component of the disease. The risk ratio, labeled λ_R , where R is the type of relative (such as sibling), is the ratio of the risk of disease in the relative of type R to the prevalence of the disease in the general population. Thus in cystic fibrosis, $\lambda_s = 1/4 \div 1/1600 = 400$. This means that the risk of developing cystic fibrosis is four hundred times greater for siblings of an affected individual than for an individual in the general population. This clearly demonstrates the effect that genetics plays in the development of cystic fibrosis.

As a general rule, the larger the λ_R , the stronger the genetic component. However, this ratio is extremely sensitive to the frequency of the disease in the general population. The more common the disorder, the lower the λ_R , although this does not necessarily preclude a genetic component to the disorder.

Twin and adoption studies provide a unique opportunity to tease apart the role of genetics and the influence of a common familial environment. Because twins were born at the same time, the environments to which they were exposed are very similar. This holds true for the prenatal environment, and often for the childhood environment, but rarely for adult environments.

Monozygotic (MZ), or identical, twins share 100 percent of their genetic makeup, and dizygotic (DZ), or fraternal, twins share on average 50 percent of their genetic makeup. Twin studies compare disease “concordance” in MZ twins with DZ twins. When both twins in a pair have the disease, the twins are said to be concordant. When one twin has the disease and the other does not, the twins are said to be discordant. If the disorder has a genetic component, then MZ twins will be concordant more often than DZ twins. The difference between the MZ and DZ concordance rates can be used to assess the strength of the genetic component.

In summary, geneticists are finding that some disorders have a large genetic effect and others have a small genetic effect. There are several methods geneticists can use to determine the size of the genetic effect of a disease. However, ultimately, researchers trying to fully understand genetic disorders must identify the genes that are involved and determine their function. The revolution in human genetics, primarily due to the successes of

the Human Genome Project, has made and will continue to make an impact on scientists' ability to define the role of genetics in human disease. SEE ALSO ALZHEIMER'S DISEASE; CANCER; CARDIOVASCULAR DISEASE; COMPLEX TRAITS; CYSTIC FIBROSIS; DIABETES; EPIDEMIOLOGIST; GENE DISCOVERY; HEMOGLOBINOPATHIES; INHERITANCE PATTERNS; MUSCULAR DYSTROPHY; TRIPLET REPEAT DISEASE; TWINS.

Allison Ashley-Koch

DNA

DNA (deoxyribonucleic acid) was discovered in the late 1800s, but its role as the material of heredity was not elucidated for fifty years after that. It occupies a central and critical role in the cell as the genetic information in which all the information required to duplicate and maintain the organism. All information necessary to maintain and propagate life is contained within a linear array of four simple bases: adenine, guanine, thymine, and cytosine.

DNA was first described as a monotonously uniform helix, generally called B-DNA. However, we now know that DNA can adopt many different shapes and conformations. Moreover, many of these alternative shapes have biological importance. Thus, the DNA is not simply an informational repository, from which information flows through RNA into proteins. Rather, structural information exists within the specific sequence patterns of the bases. This structural information dictates the interaction of DNA with proteins to carry out processes of DNA replication, transcription into RNA, and repair of errors or damage to the DNA.

The Components of DNA

DNA is composed of purine (adenine and guanine) and pyrimidine (cytosine and thymine) bases, each connected through a ribose sugar to a phosphate backbone. Many variations are possible in the chemical structure of the bases and the sugar, and in the structural relationship of the base to the sugar that result in differences in helical shape and form. The most common DNA helix, B-DNA, is a double helix of two DNA strands with about 10.5 **base pairs** per helical turn.

Bases and Base Pairs. The four bases found in DNA are shown in Figures 1 and 2. The purines and pyrimidines are the informational molecules of the genetic blueprint for the cell. The two sides of the helix are held together by hydrogen bonds between base pairs. **Hydrogen bonds** are weak attractions between a hydrogen atom on one side and an oxygen or nitrogen atom on the other. Hydrogen atoms of amino groups serve as the hydrogen bond donor while the carbonyl oxygens and ring nitrogens serve as hydrogen bond acceptors. The specific location of hydrogen bond donor and acceptor groups gives the bases their specificity for hydrogen bonding in unique pairs. Thymine (T) pairs with adenine (A) through two hydrogen bonds, and cytosine (C) pairs with guanine (G) through three hydrogen bonds (Figure 2). T does not normally pair with G, nor does C normally pair with A.

base pairs two nucleotides (either DNA or RNA) linked by weak bonds

hydrogen bonds weak bonds between the H of one molecule or group and a nitrogen or oxygen of another



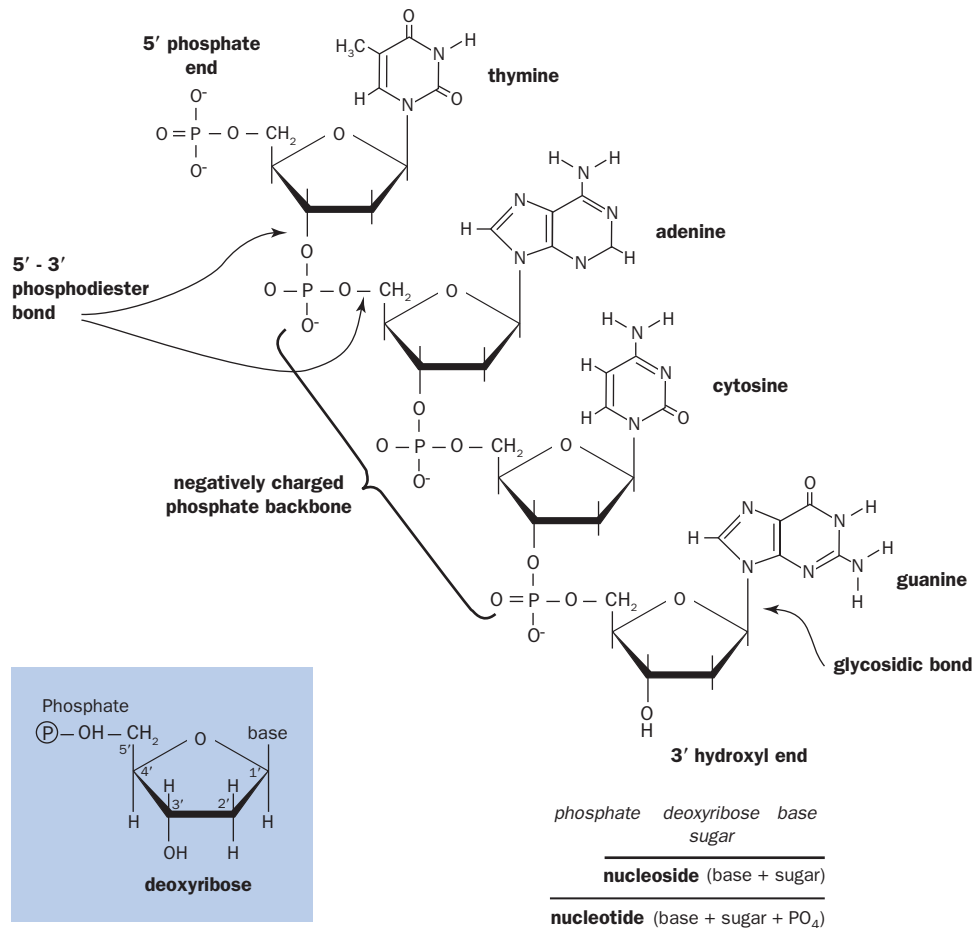


Figure 1. DNA nucleotides include a base, a deoxyribose sugar, and a phosphate. The carbons on the sugar are numbered 1' ("one prime") to 5'. The phosphate group links the 3' end of one nucleotide to the 5' end of the next. The phosphates are negatively charged, promoting their attraction to positively charged histone proteins (not shown).

β the Greek letter beta

nucleotide the building block of RNA or DNA

Deoxyribose Sugar. In DNA the bases are connected to a β -D-2-deoxyribose sugar with a hydrogen atom at the 2' ("two prime") position. The sugar is a very dynamic part of the DNA molecule. Unlike the **nucleotide** bases, which are planar and rigid, the sugar ring is easily bent and twisted into various conformations (which exist in different structural forms of DNA). In canonical B-DNA, the accepted and most common form of DNA, the sugar configuration is known as C2' endo.

Nucleosides and Nucleotides. The term "nucleoside" refers to a base and sugar. "Nucleotide," on the other hand, refers to the base, sugar, and phosphate group (Figure 1). A bond, called the glycosidic bond, holds the base to the sugar and the 3'–5' ("three prime-five prime") phosphodiester bond holds the individual nucleotides together. Nucleotides are joined from the 3' carbon of the sugar in one nucleotide to the 5' carbon of the sugar of the adjacent nucleotide. The 3' and the 5' ends are chemically very distinct and have different reactive properties. During DNA replication, new nucleotides are added only to the 3' OH end of a DNA strand. This fact has important implications for replication.

The Structure of Double-Stranded DNA

As mentioned above, the two individual strands are held together by hydrogen bonds between individual T·A and C·G base pairs. In DNA, the

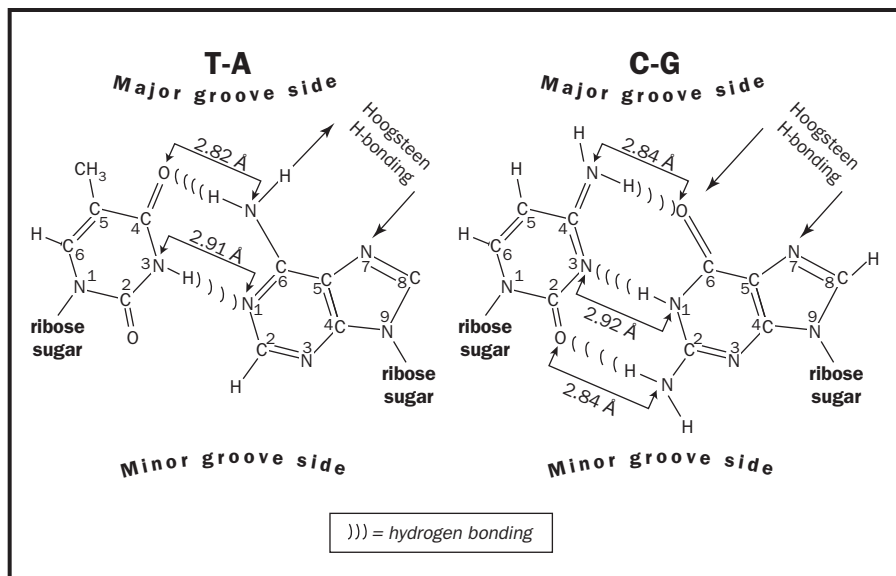


Figure 2. Hydrogen bonding between the bases. Note that the sets of distances between the base pairs are almost identical for the two base pairs, so that the distance across the double helix is unchanged. Hoogsteen hydrogen bonding can link a double helix to a third strand, to make a triple helix.

distance between the atoms involved is 2.8 to 2.95 angstroms (10^{-10} meters). While individually weak, the large number of hydrogen bonds along a DNA chain provides sufficient stability to hold the two strands together.

The stabilization of duplex (double-stranded) DNA is also dependent on base stacking. The planar, rigid bases stack on top of one another, much like a stack of coins. Since the two purine-pyrimidine pairs (A·T and C·G) have the same width, the bases stack in a rather uniform fashion. Stacking near the center of the helix affords protection from chemical and environmental attack. Both **hydrophobic interactions** and **van der Waal's forces** hold bases together in stacking interactions. About half the stability of the DNA helix comes from hydrogen bonding, while base stacking provides much of the rest.

Double-stranded DNA in its canonical B-form is a right-handed helix formed by two individual DNA strands aligned in an antiparallel fashion (a right-handed helix, when viewed on end, twists clockwise going away from the viewer). Antiparallel DNA has the two strands organized in the opposite polarity, with one strand oriented in the 5'–3' direction and the other oriented in the 3'–5' direction.

In the right-handed B-DNA double helix, the stacked base pairs are separated by about 3.24 angstroms with 10.5 base pairs forming one helical turn (360°), which is 35.7 angstroms in length. Two successive base pairs, therefore, are rotated about 34.3° with respect to each other. The width of the helix is 20 angstroms. An idealized model of the double helix is shown in Figure 3. As can be seen, the organization of the bases creates a major groove and a minor groove.

Adenine and thymine are said to be complementary, as are cytosine and guanine. Complementary means “matching opposite.” The shapes and charges of adenine and thymine complement each other, so that they attract one another and link up (as do cytosine and guanine). Indeed, one entire strand of duplex DNA is complementary to the opposing strand. During replication, the two strands unwind, and each serves as a template

hydrophobic interactions attraction between portions of a molecule based on mutual repulsion of water

van der Waal's forces weak attractive forces between two different molecules

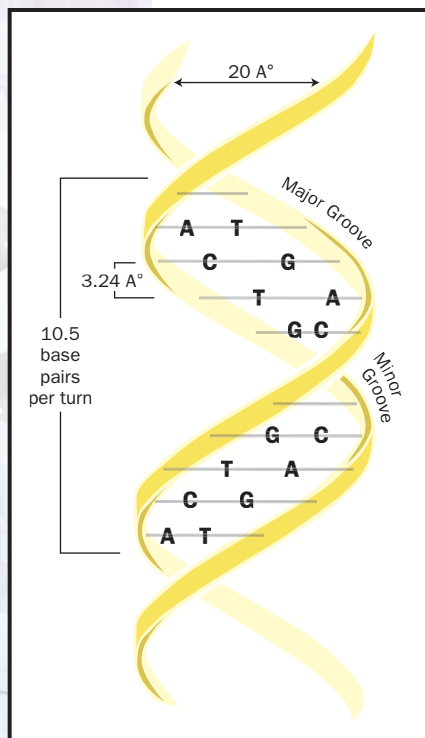


Figure 3. Canonical B-DNA double helix.

for formation of new complementary strand, so that replication ends with two exact double-stranded copies.

Alternative DNA Conformations

While the vast majority of the DNA exists in the canonical B-DNA form, DNA can adopt an amazing array of alternative structures. This is the result of certain particular sequence arrangements of DNA and, in many cases, energy in the DNA double helix from DNA supercoiling, the property of DNA in which the double helix, in a high-energy state, becomes twisted around itself. Some alternative DNA conformations identified are shown in Figure 4.

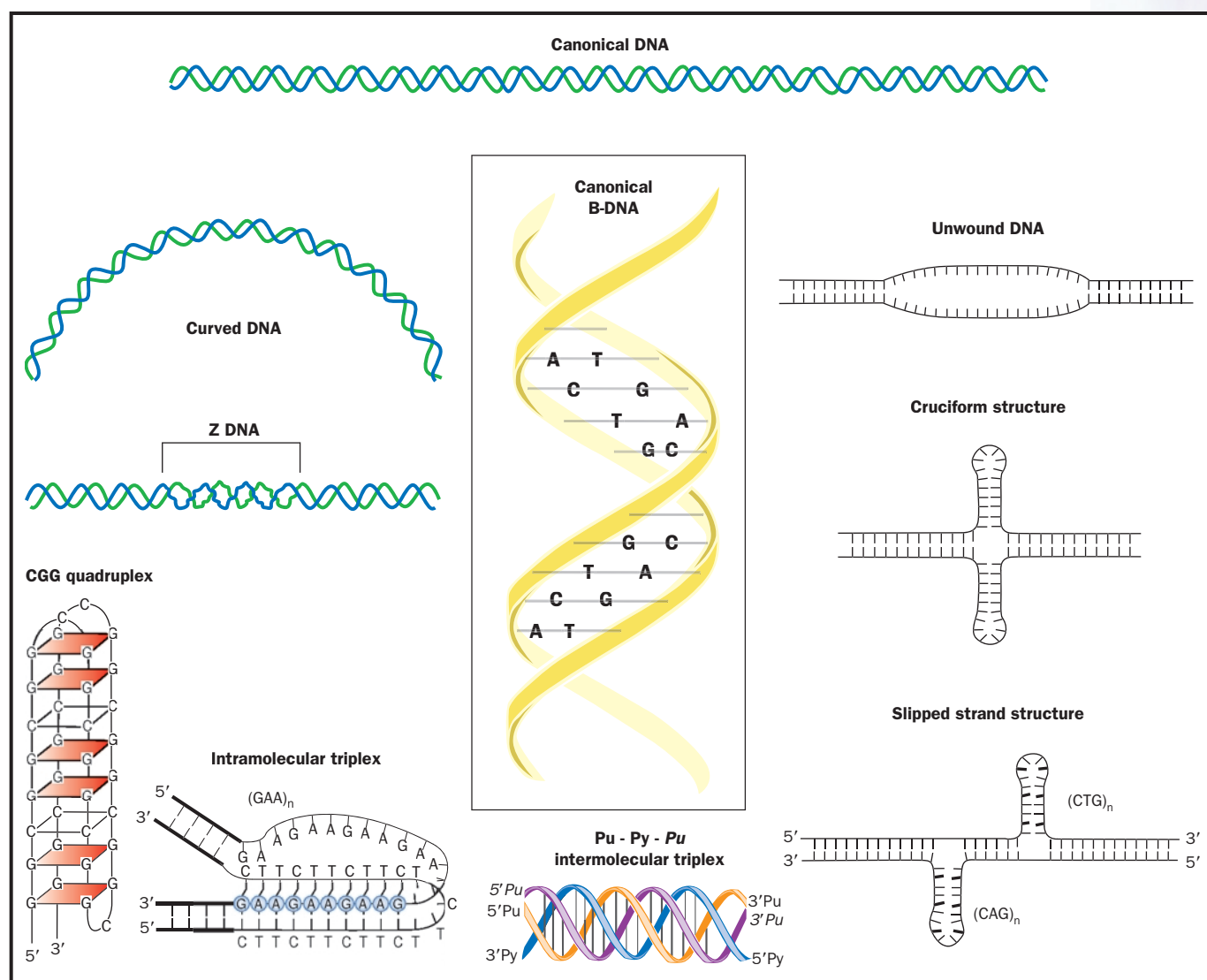
Unwound DNA. Since A·T base pairs contain two hydrogen bonds and C·G base pairs contain three, A+T-rich tracts are less thermally stable than C+G-rich tracts in DNA. Under denaturing conditions (heat or alkali), the DNA begins to “melt” (separate), and unwound regions of DNA will form, and it is the A+T-rich sequences that melt first. In addition, in the presence of superhelical energy (a high-energy state of DNA resulting from its supercoiling, which is the natural form of DNA in the chromosomes of most organisms), A+T-rich regions can unwind and remain unwound under conditions normally found in the cell. Such sites often provide places for DNA replication proteins to enter DNA to begin the process of chromosome duplication.

Cruciform Structures. DNA sequences are said to be palindromic when they contain inverted repeat symmetry, as in the sequence GGAATT·AATTCC, reading from the 5' to the 3' end. Palindromic sequences can form intramolecular bonds (within a single strand), rather than the normal intermolecular (between the two complementary strands), hydrogen bonds. To form cruciforms (“cross-shaped”), the DNA must form a small unwound structure, and then base pairs must begin to form within each individual strand, thus forming the four-stranded cruciform structure.

Slipped-Strand DNA. Slipped-strand DNA structures can form within direct repeat DNA sequences, such as $(CTG)_n \cdot (CAG)_n$ and $(CGG)_n \cdot (CCG)_n$ (where “n” denotes a variable number of repetitions). They form following denaturation, after the strands become unwound, and during renaturation, when the strands come back together. To form slipped-strand DNA, the opposite strands come together in an out-of-alignment fashion, during renaturation. Expansion of such triplet repeats are features of certain neurological diseases.

Intermolecular Triplex DNA. Three-stranded, or triplex DNA, can form within tracts of polypurine·polypyrimidine sequence, such as $(GAA)_n \cdot (TTC)_n$. Purines, with their two-ring structures, have the potential to form hydrogen bonds with a second base, even while base paired in the canonical A·T and G·C configurations. This second type of base pair is called a Hoogsteen base pair, and it can form in the major groove (the top of the base pair representations in Figure 2). Pyrimidines can only pair with a single other base, and thus a long Pu·Py tract must be present for triplex DNA formation. The important factor for triplex DNA formation is the presence of an extended purine tract in a single DNA strand. The third-strand base-pairing code is as follows: A can pair with A or T; G can pair with a **protonated C** (C^+) or G.

protonated possessing excess H^+ ions; acidic



Intramolecular Triplex DNA. When a Pu·Py tract exists that has mirror repeat symmetry (5' GAAGAG–GAGAAG 3'), an intramolecular triplex can form, in which half of the Pu·Py tract unwinds and one strand wraps into the major groove, forming a triplex. The structure in Figure 4 shows the pyrimidine strand (CTT) pairing with the purine strand (GAA) of a canonical DNA duplex. In an intramolecular triplex, one strand of the unwound region remains unpaired, as shown.

Quadruplex DNA. DNA sequences containing runs of G·C base pairs can form quadruplex, or four-stranded DNA, in which the four DNA strands are held together by Hoogsteen hydrogen bonds between all four guanines. The four guanines are aligned in a plane, and the successive rings of guanines are stacked one upon another.

Left-handed Z-DNA. Alternating runs of $(CG)_n \cdot (CG)_n$ or $(TG)_n \cdot (CA)_n$ dinucleotides in DNA, under superhelical tension or high salt (more than 3 M NaCl) (M, moles per liter) can adopt a left-handed helix called Z-DNA. In this form, the two DNA strands become wrapped in a left-handed helix, which is the opposite sense to that of canonical B-DNA. This can occur

Figure 4. DNA can exist in a variety of conformations. “Canonical” DNA is the most common, but each of the other forms have particular functions or are found in certain conditions.

within a small region of a larger right-handed B-DNA molecule, creating two junctions at the B-Z transition region.

Curved DNA. DNA containing tracts of $(A)_{3-4} \cdot (T)_{3-4}$ (that is, runs of three or four bases of A in one strand and a similar run of T in the other) spaced at 10-base pair intervals can adopt a curved helix structure.

In summary, DNA can exist in a very stable, right-handed double helix, in which the genetic information is very stable. Certain DNA sequences can also adopt alternative conformations, some of which are important regulatory signals involved in the genetic expression or replication of the DNA. SEE ALSO CHROMOSOME, EUKARYOTIC; CHROMOSOME, PROKARYOTIC; DNA MICROARRAYS; GENE; GENOME; NUCLEOTIDE; SEQUENCING DNA; TRIPLET REPEAT DISEASE.

Richard R. Sinden

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DNA Footprinting

DNA footprinting is a technique for identifying exactly where a protein binds to DNA. Knowing where a protein binds to DNA often aids in understanding how gene expression is regulated. Consequently, DNA footprinting is often part of a larger study to determine how a particular gene is controlled.

How It Works

DNA footprinting is based on the observation that when a protein binds to DNA, the DNA is protected from chemicals that would otherwise **cleave** it. In a typical DNA footprinting experiment, a DNA fragment with a suspected protein-binding site is first isolated, then labeled with a radioactive **nucleotide** or other chemical that will allow it to be detected later on.

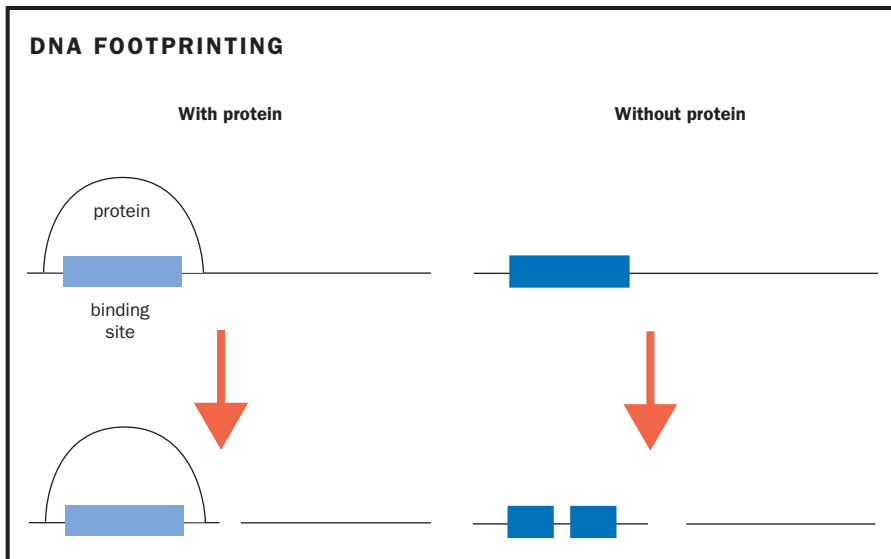
Once labeled, the DNA is then mixed in a test tube with a DNA-binding protein and a chemical that cleaves DNA, such as the **enzyme** DNase I. In a separate test tube, more of the same labeled DNA is mixed with the same cleaving chemical, but without the binding protein. The DNA fragments in each tube are allowed to incubate long enough for the molecule to cleave once, and then are separated by size (fractionated) in a DNA sequencing gel.

The reactions in the two test tubes (one with the binding protein and one without) are then compared. If the DNA actually contains protein-binding sites, these will have been protected from cleaving in the test tube that contains DNA-binding protein, and a “footprint” of those sites where no DNA cleavage occurred will be observed. By comparison with a sequencing reaction run on the same gel, one can determine the exact location where a protein has been bound to the DNA. A related technique, called gel retardation, can also be used to test for protein binding to DNA, but this method is less precise than DNA footprinting.

cleave split

nucleotide the building block of RNA or DNA

enzyme a protein that controls a reaction in a cell



The protein protects the binding site region from cleavage by DNase or another treatment.

Uses in Research

DNA footprinting is often used to locate the binding site for proteins that regulate **transcription**. For example, a researcher may suspect that a particular protein binds to a particular DNA fragment and inhibits transcription. After conducting a DNA footprinting experiment, the researcher will know the location of the exact sequence of DNA bound by that protein. If that sequence matches the sequence of a **promoter** the DNA footprinting experiment can help explain how that DNA-binding protein carries out its function.

Modified DNA footprinting experiments can also be performed to detect where proteins bind to DNA in a living cell. In these experiments, cells are grown under conditions where the protein of interest would be expected to bind to DNA. The cells are then treated with a chemical that causes proteins bound to DNA to become permanently attached to the DNA. The resulting DNA-protein complexes are then purified from the cell, and the DNA sequences are identified.

Since DNA footprinting is used to identify the specific sequences in DNA where a protein binds, the technique is likely to be of continuing usefulness in genetic research. For example, DNA footprinting is likely to be heavily used in characterizing the function of proteins identified in the Human Genome Project and other genome projects, making it an important component of a molecular geneticist's toolbox. SEE ALSO GENE EXPRESSION: OVERVIEW OF CONTROL; NUCLEASES; SEQUENCING DNA.

Patrick G. Guilfoile

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transcription messenger RNA formation from a DNA sequence

promoter DNA sequence to which RNA polymerase binds to begin transcription

DNA Libraries

DNA libraries, like conventional libraries, are used to collect and store information. In DNA libraries, the information is stored as a set of DNA molecules, each of which contains biological sequences that can be used for a variety of applications. All DNA libraries are collections of DNA fragments that represent a particular biological system of interest. By analyzing the DNA from a particular organism or tissue, researchers can answer a variety of important questions. The two most common uses for these DNA collections are DNA sequencing and gene cloning.

The Importance of Vectors

Several types of DNA libraries have been developed for specific purposes, but all share some common features. The DNA fragments that make up the library are attached to other DNA sequences that are used as “handles” to maintain the fragments. These “handles,” called vectors, allow the DNA to be replicated and stored, typically within model organisms such as yeast or bacteria.

vectors carriers

plasmid a small ring of DNA found in many bacteria

Different types of **vectors** can be used to store DNA fragments of different lengths. For example, **plasmid** vectors can store small fragments (from a few hundred bases up to ten or twenty thousand bases of sequence), while viral vectors, or viral-plasmid hybrids such as cosmids, can store up to fifty thousand bases, and yeast artificial chromosome (YAC) vectors can store hundreds of thousands of bases. In general, plasmid-based vectors are the easiest to manipulate but store the smallest fragments. They are commonly used for applications that involve complex manipulations, such as cloning or gene expression, but that require only small DNA fragments (e.g., cDNA libraries, as described below).

Artificial Chromosome Vectors and Genomic Libraries

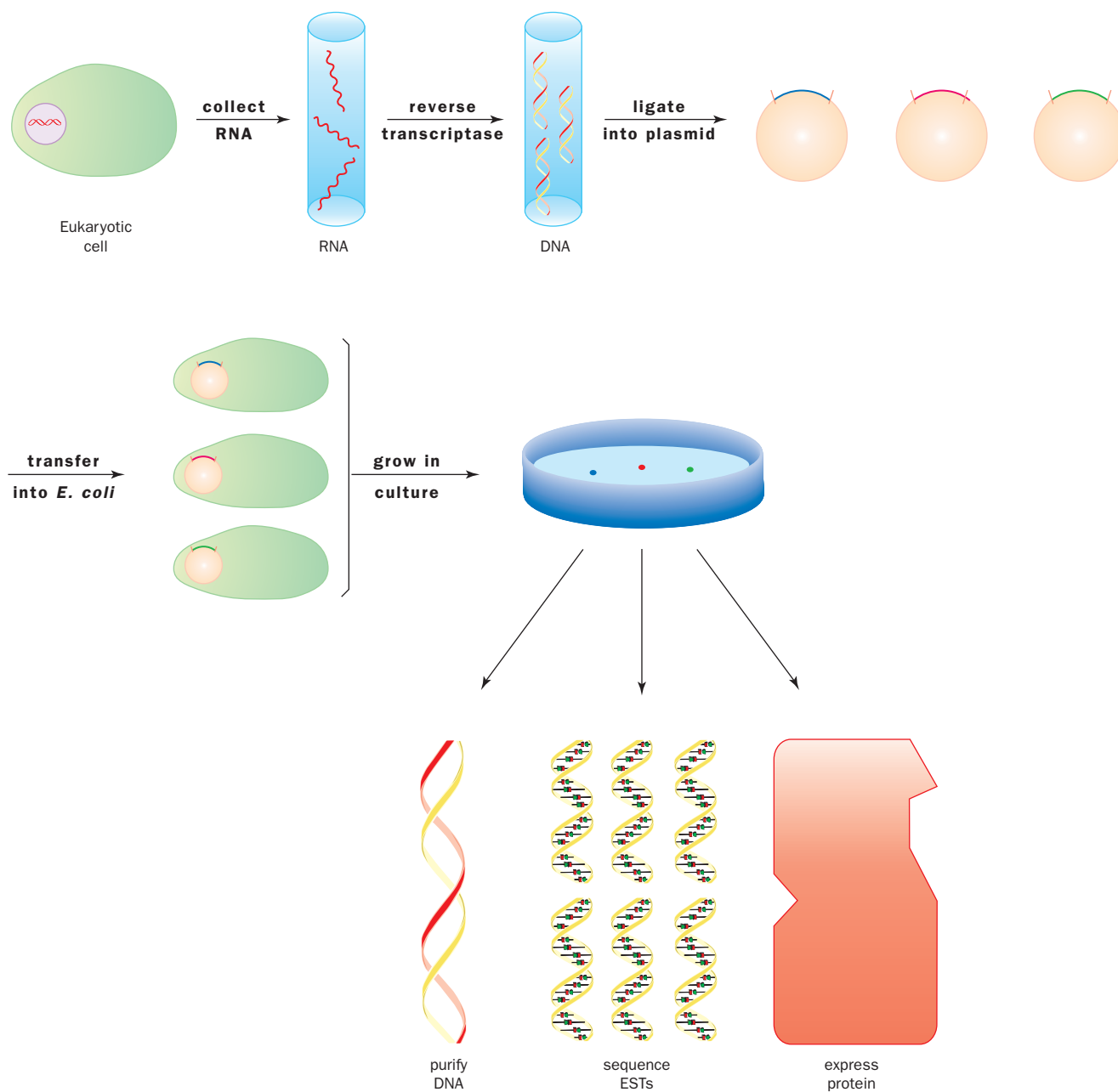
Yeast artificial chromosome vectors act like real chromosomes in yeast and can store much longer DNA fragments, some over 150 kilobases in size, big enough for several genes along with their regulatory sequences. However, YAC vectors are difficult to manipulate, are prone to spontaneous rearrangement, and have been supplanted by bacterial artificial chromosome (BAC) vectors.

Escherichia coli
common bacterium of the human gut, used in research as a model organism

BAC vectors are derived from the F plasmid of *Escherichia coli*. This plasmid behaves like a chromosome and not like a typical plasmid. BACs can store very large DNA fragments—in excess of three hundred kilobases in some cases, although typical fragments are about half that size. The unique features of BAC vectors are very well suited to creating and maintaining DNA libraries. For example, once a BAC vector enters a cell, it will exclude all other BAC vectors, which means that a given *E. coli* clone will contain only one unique library fragment. Furthermore, *E. coli* cells are relatively easy to grow and store, and DNA purification from the bacterium is straightforward. BAC libraries played a key role in the massive sequencing efforts that made up the Human Genome Project.

Many of the large-format DNA libraries (YACs, BACs) are used exclusively to store genomic DNA for sequencing projects. Larger fragments per-

CREATING A cDNA LIBRARY



A cDNA (complementary DNA) library begins with all the RNA expressed by a cell under a specific condition. The RNA is reverse transcribed into DNA, and then each DNA copy is inserted into a separate plasmid. These are taken up by *E. coli* bacteria, one to a cell, which are then grown in culture. At that point, the DNA can be purified for analysis or further cloning, “expressed sequence tags” for each can be sequenced, or the proteins can be re-expressed by the bacteria to yield enough protein for further study.

mit easier assembly of finished DNA sequence and require the maintenance of fewer clones, which is particularly important when sequencing large genomes. DNA sequencing, however, is only one application of DNA libraries.

DNA libraries are created by generating a set of DNA fragments of the desired size and then attaching those fragments to the appropriate vector sequence. For genomic DNA, the fragments are normally generated by either enzymatic digestion or simple mechanical shearing of all the DNA of the genome, including noncoding sequences. Fragments are then enzymatically attached to the vector sequences, in a reaction known as ligation. The collected fragments, now attached to vector sequences, are then moved into the appropriate host organism for growth and evaluation. Conditions are chosen so that only one fragment enters each organism, which can then be grown up into a colony whose individuals all carry the same fragment.

Complementary DNA Libraries

Genomic DNA is not always the source of the fragments in a DNA library. A second major class of libraries uses **cDNA** which is generated by copying the messenger **RNA** from an organism or tissue of interest. Because it reflects the mRNA content of a biological system (or cell type) at a particular time and under particular conditions, a cDNA library can be considered a “snapshot” of gene expression in that system. This information can be of great value in understanding when and how certain genes are expressed in an organism or cell type. Additionally, cDNA, unlike genomic DNA, lacks **introns** and other noncoding segments of sequence and is relatively straightforward to clone and express. This greatly facilitates the analysis of gene products (proteins) in **eukaryotes**.

Creating a cDNA library is similar to creating a genomic DNA library, except that the starting material for cDNA libraries is mRNA, not DNA. The enzyme reverse transcriptase is used to copy the mRNA to DNA. The DNA fragments are then cloned into vectors (typically plasmids) by ligation and moved into a host organism, as with genomic libraries.

Often, cDNA libraries are constructed using plasmid vectors with sequences that allow the cloned cDNA fragments to be expressed as proteins. Such “expression libraries” can be searched with protein-finding tools such as antibodies, and then the gene coding for the protein can be isolated. cDNA libraries are also used for expressed sequence tag (EST) analysis, in which small portions of many cDNAs are sequenced to provide an overview of gene expression in a particular sample.

DNA libraries play important roles in modern molecular biology research. The many genome-sequencing projects that are revolutionizing our understanding of genetics are entirely dependent on genomic DNA library techniques. cDNA libraries are invaluable in the study of gene expression and protein function, and for EST analysis. Continued progress in the development of library techniques and a continued interest in their applications suggest that these tools will remain an important part of the field for years to come. SEE ALSO CHROMOSOMES, ARTIFICIAL; CLONING GENES; MODEL ORGANISMS; PLASMID; POLYMERASE CHAIN REACTION; RESTRICTION ENZYMES; REVERSE TRANSCRIPTASE.

cDNA complementary DNA

RNA ribonucleic acid

introns untranslated portions of genes which interrupt coding regions

eukaryotes organisms with cells possessing a nucleus

Daniel J. Tomso

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DNA Microarrays

DNA microarrays are tools used to analyze and measure the activity of genes. Researchers can use microarrays and other methods to measure changes in gene expression and thereby learn how cells respond to a disease or to some other challenge.

Gene Expression

Humans have 30,000 to 70,000 genes, each consisting of a sequence of bases, the building blocks of the hereditary material DNA. Before they can carry out their function, genes are copied to make messenger RNA (mRNA), in a process called **transcription**. This molecule is in turn used as a template for the synthesis of a protein molecule (**translation**). This entire process, including transcription of RNA and translation of protein, is referred to as gene expression. Only a subset of the full set of genes is expressed in a given tissue at a given time. In fact, this differential pattern of gene expression is ultimately what distinguishes lung tissue from skin, liver, and muscle tissue.

Even within a given tissue type, different genes are expressed at different times. For example, there is a very tightly controlled sequence of gene expression during the course of embryonic development. Tissues also respond to metabolic and other challenges. The pattern of gene expression changes in the liver in response to the consumption of a large meal. Similarly, muscle gene expression changes in response to vigorous exercise or injury. Drugs can also affect gene expression. Researchers can use microarrays and other methods to measure these changes in gene expression, and from them learn about how cells respond to disease or to other challenges.

Hybridization

Microarrays measure gene expression by taking advantage of the process of **hybridization (molecular)**. DNA is made up of four bases: guanine, adenine, cytosine, and thymine, which are abbreviated G, A, C, and T, respectively. G and C can bind to one another, forming a base pair, as can A and T, but no other combinations of bases can form base pairs. G and C are said to be “**complementary**” bases, as are A and T.

The bases on each of the two strands of DNA that make up a chromosome are complementary to the bases on the opposite strand. Long pieces of DNA will not bind to each other (or “hybridize”) unless they are complementary. Hybridization allows researchers to test whether two pieces of DNA are complementary. If they bind to one another (hybridize) then they are opposite strands of a single gene. If they do not bind to one another, then they are unrelated.

Hybridization can be used to measure the levels of hundreds of different mRNAs within a given tissue, thereby providing a picture of gene

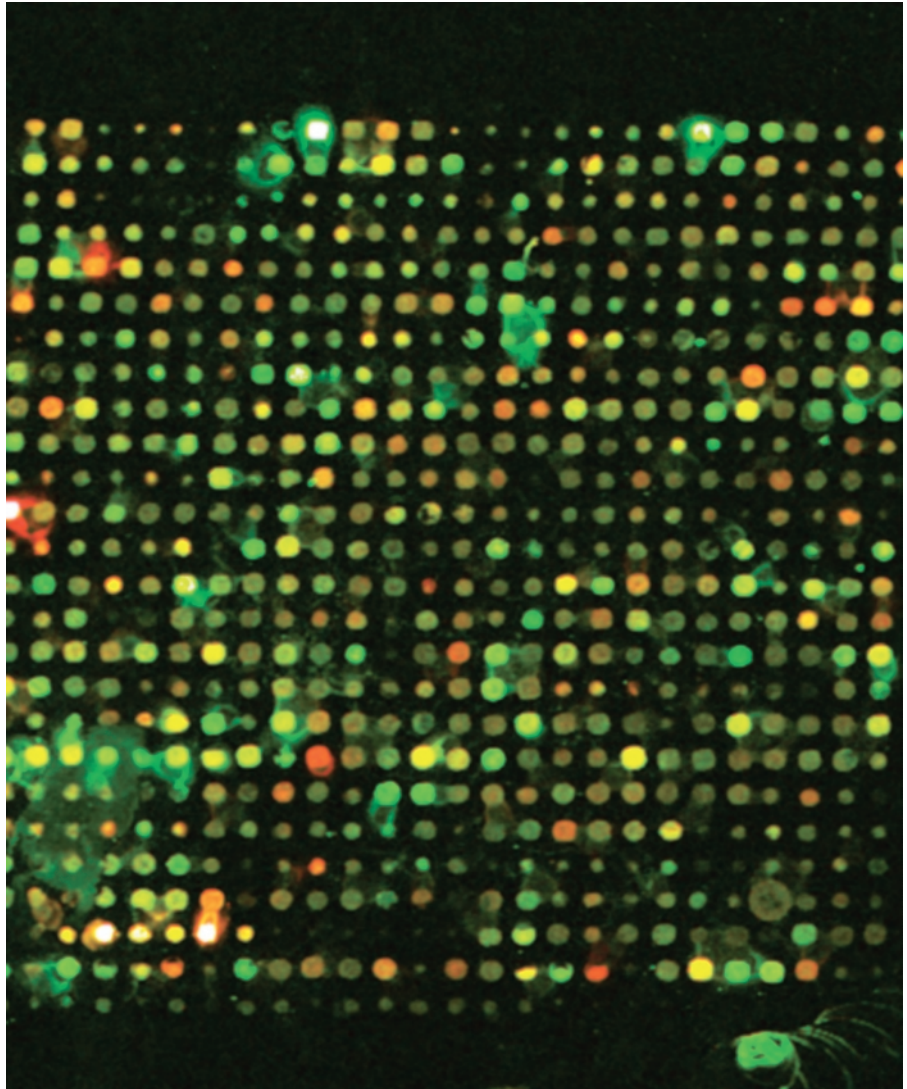
transcription messenger RNA formation from a DNA sequence

translation synthesis of protein using mRNA code

hybridization (molecular) base-pairing among DNAs or RNAs of different origins

complementary matching opposite, like hand and glove

Tools like this DNA microarray aid researchers in measuring gene expression. Each color represents a different level of gene expression.



expression within that tissue. RNA is isolated from the tissue of interest and allowed to hybridize to a solid support to which many different DNA pieces, from many different genes, have been attached. Because the RNA is labeled with a fluorescent tag, the amount bound to a given spot can be measured. The fluorescent intensity of each spot is a measure of the level of that mRNA that was expressed in the original tissue. In this way, the levels of expression of up to 12,000 different genes can be measured with a single microarray.

There are two basic types of microarrays. One type is created by a company called Affymetrix. Affymetrix manufactures silicon and glass chips that resemble semiconductor chips and that are manufactured using the same photolithographic techniques. These chips have sets of very short (20 base-pair) stretches of DNA representing each gene. A second type of microarray is commonly called a printed array and is made by spotting small amounts of DNA on glass slides. These arrays frequently have smaller numbers of genes on each slide, but researchers can easily modify them for specific experiments.

Microarray Analysis

Microarrays produce enormous amounts of data, and the analysis of that data can be quite complex. The sheer volume of data requires special software and a database in which to store both the measurements and the results of the analyses. The exact form that the analysis takes depends on the nature of the experiment being performed. If just two samples are being directly compared (for example, gene expression in mouse heart tissue is compared with and without the administration of a drug), relatively straightforward statistical tests can be performed. If larger numbers of samples are being measured, the same tests can be performed between two samples at a time, but more sophisticated, “clustering” analyses can be performed as well.

Clustering analysis identifies groups of genes that react the same way across several different samples. For example, researchers might analyze gene expression in heart tissue from a set of mouse embryos that range in age from five to fifteen days. A clustering analysis would be able to detect a group of genes whose expression levels all increase slowly from days five to nine, peak at day ten and then fall to zero by day twelve. Only genes that have this precise pattern of expression would cluster together, in this type of analysis.

The Role of Bioinformatics

One of the tremendous difficulties in performing any kind of expression analysis is the manipulation of very large amounts of biological data, a field of study called bioinformatics. The usefulness of gene expression data depends on how much information is available for each identified gene. In other words, the identities of the genes associated with each spot on a microarray must be accessible as the analysis is done.

Descriptions and classifications of each gene on the array must be readily available, as no researcher can remember such details about the tens of thousands of genes that may be involved in the analysis. An analysis might be done many times, with slight changes in the parameters of the clustering algorithm each time. The genes that cluster together are examined at the end of each analysis, to look for reproducible patterns. This analysis must be done with the full understanding of the biology of the system being studied. Clusters of genes are most informative if they group in a biologically reasonable way. For this reason, microarray expression analysis is frequently exploratory. The results of the analysis are used to suggest additional, corroborative experiments.

Another bioinformatics challenge in gene expression studies is collecting information about the samples under analysis and storing the information in databases. If gene expression patterns of one hundred different **tumor** samples are being examined, it may be necessary to restrict the analysis to subgroups of the tumors in order to observe patterns in the data. This subgrouping or stratification of the samples is best performed on the basis of independently determined properties of those samples. For example, samples from only **metastatic** cancer cells could be grouped together for analysis and compared with those from nonmetastatic cancer cells, or the age of the patient at the onset of disease could be used to segregate the samples into different groups. Such subgroup analysis can only be done if complete information is collected and stored for all samples.

tumor mass of undifferentiated cells; may become cancerous

metastatic broken away cancerous cells from the initial tumor

linkage analysis examination of co-inheritance of disease and DNA markers, used to locate disease genes

expression analysis whole-cell analysis of gene expression (use of a gene to create its RNA or protein product)

Applications of Microarray Analysis

Microarrays are new enough that their applications are still being developed. Microarray expression analysis can be used to help study complex, multigenic diseases such as Parkinson's disease (PD). The great challenge in understanding the genetics of such disorders is identifying susceptibility genes, which are genes that increase a person's risk of developing the disease. Frequently, the first step in discovering a susceptibility gene is **linkage analysis**. This technique can identify regions of a chromosome that harbor such a gene, but the regions that are identified are frequently very large, containing hundreds of genes. Screening through all of these genes individually is tremendously slow and labor-intensive. **Expression analysis** using microarrays can help prioritize these genes for further analysis by providing independent lines of evidence that specific genes are involved in the disease process.

Brain tissue can be collected through anatomical donations from patients with Parkinson's disease and from unaffected individuals, for example. Regions of the brain that are especially affected in Parkinson's patients can be compared to the same regions from unaffected individuals. Genes whose levels of expression vary can be identified. Hundreds or thousands of genes may be identified in this way, but they can then be compared to those that are found, through linkage analysis, to be linked to Parkinson's disease. There may be only tens of genes common to both groups. These genes can be prioritized for detailed examination through other methods. The key here is that expression analysis and linkage analysis provide independent evidence of a given gene's involvement in a disease process. It is the synthesis of information from these two independent lines of evidence that makes this approach powerful.

Another very powerful application of microarray expression data is called classification analysis. This technique uses gene expression data to separate tissue samples into two or more groups. For example, one type of tumor may respond very well to an aggressive program of chemotherapy treatment, while another type may respond better to surgical removal followed by radiation therapy. Further, these two types of tumors may be difficult or impossible to tell apart under a microscope. Choosing the correct method of treatment and applying that treatment early in the course of disease could significantly improve a patient's chances of survival.

In such a case, expression analysis can be used to give a detailed picture of the genes that are expressed in the two types of tumor. A training set (a small set of samples in each category) can be used to find specific patterns of gene expression that are characteristic for each type of tumor. New tumors can then be analyzed, and their expression profiles can be used to predict the group to which they belong. These approaches are used with great success to refine the clinical management of cancer patients. A 2001 study by S. Dhanasekaran, "Delineation of Prognostic Biomarkers in Prostate Cancer," offers an example of this kind of work. Additional applications of microarrays are still being developed.

SAGE Analysis

Gene expression analysis can also be done using a powerful technique called serial analysis of gene expression (SAGE). Like microarrays, SAGE starts

by isolating RNA from the tissue of interest. This RNA is then processed through a long series of steps resulting in the isolation of a set of very short sequences, called tags, from each **transcript** in the cell. These tags are converted into corresponding segments of DNA. These pieces of DNA, which are 14 base pairs long, are then linked together into long chains, and their sequence of bases is determined. Tens of thousands of these SAGE tags are sequenced from each tissue that is being studied. The tags corresponding to a given gene from one tissue are counted and compared to those from the same gene in another tissue.

For example, a colon cancer tumor sample might generate 50,000 SAGE tags, thirty-three of which correspond to a specific gene. A second library made from normal colon cells might have fifty thousand tags, eleven of which correspond to the same gene. This would indicate that the gene is expressed at a level that is three times as great in tumor cells than it is in normal cells.

SAGE data is significantly more difficult and expensive to produce than microarray data, but it offers the advantage of providing very precise and quantitative measurements of expression levels. SAGE has the further advantage that it can detect genes that have not been previously characterized. Such unknown genes cannot be detected by microarrays, because researchers must first know their sequence before they can place them on the array. SAGE therefore can be used as a gene discovery tool.

SAGE has been used most extensively in cancer research. Investigators in the Cancer Genome and Anatomy Project have created more than one hundred SAGE libraries from normal and cancerous tissue. Analysis of these libraries has revealed a great deal about the way that gene expression changes in cancerous tissue, which in turn has provided insight into new diagnostic and treatment options.

SAGE has also been used as a tool to help calculate the total number of genes in the human body, as well as to describe the ways in which genes are regulated and processed at different times. Microarrays and SAGE analysis are only two of the many ways that scientists have examined gene expression. As these techniques become more refined, and as new techniques are developed, they will provide a powerful tool to investigate how the incredible diversity and complexity of our tissues can arise, even though every cell in our bodies contains exactly the same set of genes. **SEE ALSO** BIOINFORMATICS; CANCER; COMPLEX TRAITS; GENE DISCOVERY; *IN SITU* HYBRIDIZATION; LINKAGE AND RECOMBINATION; MAPPING.

Michael A. Hauser

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transcript RNA copy of a gene



DNA Polymerases

polymerization linking together of similar parts to form a polymer

replication duplication of DNA

mutations changes in DNA sequences

complementary matching opposite, like hand and glove

nucleoside triphosphate building block of DNA or RNA, composed of a base and a sugar linked to three phosphates

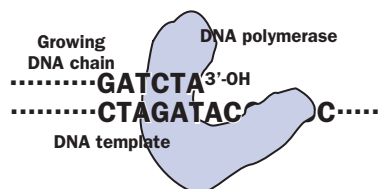
deoxynucleotides building blocks of DNA

DNA polymerases are proteins that synthesize new DNA strands using pre-existing DNA strands as templates. Before one cell divides to produce two cells, the DNA containing the genetic information in it must be duplicated for the new cell, in a process known as **polymerization**. In human cells, duplicating the DNA genome requires the polymerization of 2.91 billion nucleotides, the building blocks of DNA. In the bacterium *Escherichia coli*, the polymerization of 4.64 million nucleotides is necessary to duplicate the genome for the new cell. In all cells, the DNA polymerases are the protein catalysts that link together the nucleotide building blocks of the new DNA polymer in an accurate and timely process that occurs during **replication**.

The DNA polymerases are also required to repair the DNA of the genome. The genome's DNA can be damaged by highly reactive molecules that are either produced in the cell during normal metabolic processes or brought into the cell from external sources. The damage, if not repaired, could result in the production of **mutations** in the genome or possibly cell death. Several DNA repair processes occurring in the cell have been identified that preserve the integrity of the genome by removing the damaged nucleotides and resynthesizing DNA by the DNA polymerases.

The DNA Polymerase Mechanism

All DNA polymerases share a common mechanism for DNA chain synthesis. The polymerization of DNA occurs by the linkage of one nucleotide at a time to the end of a preexisting DNA chain. The sequence fluctuations of the nucleotides on the DNA template upon which the DNA polymerase is moving determines which nucleotide is added onto the end of the growing DNA chain. If a thymine (T) nucleotide is positioned in the DNA template, for example, then an adenine (A) is polymerized onto the DNA chain opposite the thymine in the DNA template. If a guanine (G) nucleotide is positioned in the template, a cytosine (C) is linked to the growing DNA chain opposite the guanine. This polymerization process results in the synthesis of a DNA chain that is **complementary**, rather than identical, to the template strand of DNA, and is sequenced according to the proper Watson-Crick nucleotide base pairing rules. During replication, both strands of the duplex DNA molecule serve as templates. The DNA strands are separated, and each of the DNA strands is copied by the DNA polymerases. This process results in two identical copies of the original duplex DNA molecule being produced for the two cells.



The DNA polymerase uses the **nucleoside triphosphate** form of the **deoxynucleotides** to build the DNA polymer. The monophosphate form of the deoxynucleotide is incorporated into the growing DNA chain, and a pyrophosphate molecule, a kind of salt, is released. The DNA polymerase can add nucleotides only to the 3'-OH end of the growing DNA chain (see